Empowering Students with predictive Intelligence:
A Career guidance System for Sri Lanka

Group ID: 25-26J-336

Proposal Report Draft

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B.Sc. (Hons) in Information Technology Specializing in Information Technology

Sri Lanka Institute of Information Technology

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AI-Powered Soft Skill Development System for Career Readiness of Sri Lankan Students

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text

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ABSTRACT

In today's competitive world, academic achievements alone are no longer enough to ensure career success. Employers increasingly value soft skills such as communication, problem-solving, and critical thinking, yet many students struggle to develop these skills within traditional education systems that prioritize examinations over holistic growth. This gap is particularly visible in Sri Lanka, where students transitioning from O/L to A/L, A/L to university, and eventually into the professional environment often face challenges in expressing ideas clearly, reasoning effectively, and adapting to new situations.

This project proposes the design and development of an AI-powered Soft Skill Development System aimed at helping students strengthen these essential abilities. The system will evaluate a student's current skill level through speech-based tasks, analyzed using Automatic Speech Recognition (ASR) and Natural Language Processing (NLP), alongside scenario-based multiple-choice questions for problem-solving and critical thinking. Based on the results, the system will provide personalized recommendations drawn from a curated set of online courses, blogs, and tasks. A student dashboard will track progress over time, offering clear insights into improvement.

By integrating AI assessment, adaptive feedback, and personalized learning pathways, this project seeks to bridge the gap between academic knowledge and employability, equipping students with the skills needed for lifelong success..

ABBREVIATIONS

Abbreviation	**Description**
ROI	Return on investment
AI	Artificial Intelligence
ML	Machine Learning
ASR	Automatic Speech Recognition
NLP	Natural Language Processing
A/L	Advanced Level
O/L	Ordinary Level
MOOC	Massive Open Online Course

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1. INTRODUCTION

Soft skills have become a decisive factor in determining career readiness in the 21st century. While academic achievements lay the foundation for knowledge, it is the ability to communicate effectively, solve problems logically, and think critically that allows individuals to thrive in professional and personal contexts. Employers across industries consistently highlight that graduates often enter the workforce with strong technical knowledge but inadequate soft skills, limiting their ability to collaborate, innovate, and adapt to complex challenges. This mismatch between education outcomes and workplace expectations has resulted in an increasing demand for interventions that strengthen soft skills at earlier stages of learning.

In Sri Lanka, the problem is particularly evident during key educational transitions: from O/L to (A/L), A/L to university, and university to professional life. Students navigating these phases are often guided by examination-oriented curriculum that leave limited room for soft skill development. As a result, many face difficulties in articulating ideas clearly, approaching problems with structured reasoning, or demonstrating critical judgment all of which are vital in modern academic and career environments [1].

This project aims to address this gap through the design of an AI-driven Soft Skill Development System. The system focuses on three core areas: communication, problem-solving, and critical thinking [2]. Using speech-based evaluations processed with ASR and NLP, along with scenario-based multiple-choice assessments, the system will provide an objective measure of a student's current abilities. Based on these results, students will receive tailored recommendations from a curated collection of courses, articles, and activities. Progress will be visualized through a student dashboard, enabling continuous improvement. By integrating intelligent assessment with adaptive learning pathways, this project seeks to complement traditional education with a scalable, technology-driven solution for career readiness.

2. BACKGROUND

The importance of soft skills in education and employment has been widely acknowledged in recent years. While technical knowledge and subject expertise are critical, they alone are insufficient for success in today's rapidly evolving workplace [1]. Skills such as communication, problem-solving, and critical thinking play a crucial role in shaping an individual's ability to collaborate, adapt, and contribute effectively to both academic and professional environments. International research shows that graduates who demonstrate strong soft skills are more employable and better positioned to succeed in dynamic, team-oriented workplaces.

In Sri Lanka, however, the development of these skills remains limited due to the examination-focused nature of the education system. The academic journey from O/L to A/L, A/L to university, and eventually into professional life, emphasizes rote learning and test performance, leaving little space for cultivating practical abilities. Students often lack structured opportunities to practice articulating ideas, analyzing problems critically, or working through real-world scenarios that demand logical reasoning. As a result, many struggle with soft skills when transitioning into university group projects, presentations, or professional interviews, which contributes to employability challenges and underutilization of graduate talent [2].

Technology presents an opportunity to bridge this gap by offering scalable and adaptive solutions that can complement existing education. AI particularly tools like ASR and NLP [3], enables objective assessment of communication, while structured tasks and quizzes provide insights into problem-solving and critical thinking. By integrating these approaches with personalized recommendations and progress tracking, an AI-driven system can help Sri Lankan students strengthen essential soft skills and better prepare for future academic and career challenges.

3. LITERATURE SURVEY

A. Soft Skill Evaluation Function

The evaluation of soft skills remains a complex task, as attributes such as communication, problem-solving, and critical thinking are less tangible compared to technical knowledge. Traditional assessment methods rely on self-report surveys and teacher evaluations, which, while valuable, often suffer from subjectivity and bias. Recent advancements in Artificial Intelligence have introduced more objective mechanisms. ASR and NLP [4] have been successfully applied to analyze verbal communication, measuring aspects such as fluency, coherence, and vocabulary richness. Similarly, scenario-based quizzes and structured problem-solving tasks have been used to assess reasoning and decision-making skills, enabling a more reliable baseline for identifying student strengths and weaknesses.

B. Soft Skill Enhancement Function

Once skill levels are identified, the next step involves structured enhancement. Digital learning platforms increasingly use tasks, and adaptive quizzes to build soft skills in learners. For communication, interventions such as interactive speaking exercises, debate simulations, and speech analysis feedback have proven effective. Problem-solving and critical thinking are often enhanced through case studies, logic-based tasks, and puzzles designed to mirror real-world challenges. Importantly, research highlights that enhancement strategies are most effective when they are personalized and context-aware, aligning with a student's current skill profile rather than applying a one-size-fits-all approach [6].

C. Soft Skill Progress Tracking

Recommendation systems have been widely studied in the educational domain to guide learners toward relevant resources. Traditional systems use content-based or collaborative filtering, but recent studies show the value of hybrid and adaptive approaches (LightFM, TensorFlow Recommenders) [7]. In the context of soft skills, researchers have experimented with recommending MOOCs, articles, and workshops that directly target areas of weakness identified during assessment [5]. While some projects rely on dynamic content fetched from external APIs, many adopt a curated list of high-quality resources for feasibility and reliability, especially in academic prototypes. Personalized recommendations not only promote targeted growth but also increase student engagement by providing actionable pathways for improvement.

D. Soft Skill recommendation

Feedback plays a critical role in sustaining student motivation and ensuring long-term improvement. Studies on AI-driven learning systems emphasize the importance of timely, constructive, and actionable feedback. For communication skills, systems provide insights such as “frequent pauses reduce clarity” or “positive sentiment detected in speech,” while for problem-solving, feedback may highlight gaps in logic or reasoning patterns. Visual dashboards have also gained prominence, offering learners a clear view of their progress through radar charts and performance graphs [8]. Importantly, feedback loops that integrate both automated analysis and user reflections have been found to increase student trust in AI-based systems and enhance adoption [9].

4. RESEARCH GAP

Although soft skills are increasingly recognized as critical to career readiness, existing research reveals several limitations in how these skills are assessed, enhanced, and supported through technology. Current evaluation methods often rely on self-report surveys and teacher observations, which are valuable but prone to bias and subjectivity. While some recent studies have employed ASR and NLP for communication analysis [10], their application is mostly limited to language learning or workplace training rather than targeting students in secondary and tertiary education. Similarly, problem-solving and critical thinking assessments are frequently implemented in isolated experiments, without being embedded into a continuous development framework.

On the enhancement side, most platforms adopt generic content delivery, offering the same learning materials to all students regardless of their current skill profile. This one-size-fits-all approach overlooks the importance of personalized, adaptive learning pathways. Recommendation systems in education have shown promise, but many rely on external APIs or MOOCs [11], which may not always be contextually relevant, sustainable, or feasible for resource-constrained environments like Sri Lanka.

Feedback mechanisms also present gaps: while dashboards and automated scoring exist, meaningful, actionable feedback tailored to individual learners is often missing [12]. Moreover, very few systems integrate evaluation, enhancement, recommendation, and feedback into a single loop that supports ongoing improvement.

These gaps underline the need for a comprehensive, AI-powered soft skill development system that is student-centered, phase-specific, and designed to complement Sri Lanka's examination-oriented education system by offering objective assessment, personalized enhancement, and actionable feedback

5. RESEARCH PROBLEM

Across the globe, the importance of soft skills has grown significantly, with employers consistently ranking communication, problem-solving, and critical thinking among the most desired attributes in graduates [13]. In Sri Lanka, however, the education system remains largely exam-oriented, emphasizing subject knowledge and memorization while providing few opportunities for structured soft skill development. This creates a persistent mismatch between academic outcomes and workplace expectations [14]. Students transitioning from O/L to A/L, from A/L to university, and from university to the professional world often struggle to articulate ideas clearly, reason through complex challenges, or demonstrate critical analysis. These shortcomings contribute to underemployment, dissatisfaction, and reduced career readiness.

Although research has explored AI-driven methods for evaluating and enhancing soft skills, most systems are designed for either language learning or workplace contexts [15]. Very few address the specific needs of students navigating academic transitions. Current solutions also tend to isolate evaluation from enhancement, with limited personalization in the form of recommendations or feedback. In addition, most lack cultural relevance and adaptability to local educational environments.

The problem, therefore, lies in the absence of a comprehensive, AI-powered system tailored to Sri Lankan students that can evaluate communication, problem-solving, and critical thinking objectively, recommend personalized resources for improvement, and track progress over time. Addressing this problem requires the integration of AI techniques such as ASR, NLP, and adaptive recommendation mechanisms into a scalable, student-centered module for soft skill development.

6. OBJECTIVE

6.1. Main Objectives

The main objective of this project is to design and develop an AI-powered Soft Skill Development System that evaluates, enhances, and tracks the progress of students' soft skills specifically communication, problem-solving, and critical thinking, across key educational transitions in Sri Lanka, O/L to A/L, A/L to university, and university to career. By integrating ASR, NLP, structured assessments, and personalized recommendations, the system aims to bridge the gap between traditional academic learning and essential career readiness

- To evaluate students' current soft skill levels using phase-specific surveys, scenario-based quizzes, and speech analysis via NLP tools (Whisper, spaCy, BERT), identifying strengths and gaps through machine learning models.
- To develop personalized recommendations for skill improvement, including curated resources (e.g., online courses, local workshops) and adaptive tasks, delivered via a recommendation engine tailored to individual profiles and socio-economic contexts.
- To monitor and track student progress through engagement metrics and visual dashboards, employing reinforcement learning for dynamic adaptation of recommendations to ensure continuous skill evolution.
- To pilot test the module with diverse student groups, refine based on feedback, and deploy it within a scalable web application, ensuring accessibility and integration with the broader career guidance system.

6.2. Specific Objectives

1. To evaluate students' current soft skill levels through:
    - Speech-based assessments analyzed using ASR and NLP for communication.

- Multiple-choice and scenario-based quizzes for problem-solving and critical thinking.

2. To provide targeted recommendations for improvement using a pre-curated database of courses, blogs, and articles, mapped to skill gaps identified during assessment.
3. To track and visualize progress through an interactive student dashboard that displays historical performance, skill growth trends, and recommended learning pathways.
4. To ensure adaptability of the system by adjusting tasks and recommendations according to each student's performance, enabling continuous improvement rather than one-time evaluation.
5. To test and validate the system with student groups, analyzing its effectiveness in providing accurate assessment, relevant recommendations, and meaningful progress tracking.

7. METHODOLOGY

The project will be implemented through a phased development cycle involving data collection, module design, AI/ML integration, user interface development, testing, and evaluation. A bottom-up approach will be followed starting with core functionality such as evaluation modules and building up to recommendation and tracking features. An iterative model will be used, with feedback loops embedded after each phase for refinement.

1. Data Collection

To build a reliable dataset, both speech recordings and quiz responses will be collected from students across the O/L to A/L, A/L to university, and university to career transition phases.

- **Speech Data:** Students will be asked to record short responses to predefined prompts, such as "Describe a time when you solved a problem in a group." These recordings will be used to assess communication skills.
- **Quizzes:** Scenario-based multiple-choice questions will be administered to evaluate problem-solving and critical thinking abilities.
- **Sample Size:** Around 50–100 participants will be recruited initially to create a prototype dataset.
- All data will be anonymized and stored securely to ensure privacy and ethical compliance

2. System Design

The system will consist of the following modules:

- Assessment Module: Processes speech recordings using ASR to convert audio into text. NLP models will then analyze fluency, coherence, vocabulary richness, and

sentiment. Quizzes will be automatically scored for problem-solving and critical thinking.

- Recommendation Module: Based on student scores, a rule-based recommendation engine will map identified weaknesses to curated resources such as online courses, blogs, and articles.
- Progress Tracking Dashboard: Provides students with visual insights (radar charts, line graphs) on their performance history, highlighting growth areas and recommended next steps.

3. Model Development

- ASR: OpenAI Whisper or wav2vec 2.0 will be used for speech-to-text conversion.
- NLP: spaCy, BERT, or sentence-transformers will be fine-tuned to analyze vocabulary richness, coherence, and sentiment in student responses.
- Evaluation Logic: MCQ results will be processed using simple classifiers to categorize problem-solving and critical thinking performance.
- Recommendation Engine: A rule-based system will allocate resources based on thresholds (e.g., Communication <50% → recommend speaking exercises and blogs)..

4. Implementation

- Frontend: A web-based interface built using React.js for accessibility and ease of use.
- Backend: FastAPI in Python to handle model inference, quiz scoring, and recommendation delivery.
- Database: PostgreSQL to store student profiles, skill scores, progress history, and recommendation records.
- Integration: APIs will connect ASR/NLP models with the system for real-time evaluation and feedback.

5. Evaluation

The system will be evaluated through both technical and user-centric measures:

- **Technical Evaluation:**
  - ASR accuracy measured by Word Error Rate (WER).
  - NLP scoring reliability compared to manual evaluations.
- **User Evaluation:**
  - Pilot testing with students to assess usability, clarity of feedback, and relevance of recommendations.
  - Pre-test and post-test analysis to measure skill improvements over time.

8. PROJECT REQUIREMENTS

8.1. Functional Requirements

These are the core functionalities the system must perform:

FR-01 User Access (Minimal)

- The system shall allow students to register, sign in, and maintain a basic profile (name, email, academic phase) so results can be saved per user.

FR-02 Assessment Flow Orchestration

- The system shall provide a guided assessment flow: instructions → speech task → MCQ quizzes → results summary.

FR-03 Speech Capture

- The system will record short student responses to predefined prompts and store the audio per attempt.

FR-04 ASR (English)

- The system shall convert recorded audio to text using an ASR model and persist the transcript with the attempt.

FR-05 NLP-Based Communication Scoring

The system shall analyze transcripts to produce communication sub-scores (e.g., fluency, coherence, vocabulary richness, sentiment/tonality) and an aggregated Communication score on a 0–100 scale.

FR-06 MCQ Engine (Problem-Solving & Critical Thinking)

- The system shall deliver timed, scenario-based MCQs, auto-grade responses, compute sub-dimension scores, and produce overall Problem-Solving and Critical Thinking scores on a 0–100 scale.

FR-07 Skill Profile & Results:

- The system shall generate a skill profile per attempt (Communication, Problem-Solving, Critical Thinking), display strengths/weaknesses, and save all results for longitudinal tracking.

FR-08 Rule-Based Recommendations (Pre-Curated List)

- The system shall map score thresholds to targeted recommendations (courses, blogs, articles, tasks) from a pre-curated catalog and present at least 2–3 items for each weak area.
- The system shall record which recommendations were shown, opened, and marked as completed; it shall capture simple student feedback (helpful/not helpful).

FR-09 Student Dashboard

- The system shall provide a personal dashboard showing latest and historical scores, trend lines per skill, current skill profile (e.g., radar), and the status of recommended items.

FR-10 Data Persistence & Privacy (Essential)

- The system shall securely store audio, transcripts, quiz answers, scores, and recommendation interactions per user, using anonymized identifiers where appropriate.

8.2. Non-Functional Requirements

- Performance: The system should process and return evaluation results within 5–10 seconds for each submission
- Scalability: Designed to support multiple users simultaneously with minimal performance degradation.
- Security and Privacy: Ensure secure login, access control, and data encryption. Comply with data privacy standards; no personal identifiers stored with responses.

- Usability: Interface should be intuitive for students with varying levels of digital literacy. All content should be accessible through standard desktop or mobile browsers.
- Reliability: System must recover gracefully from unexpected crashes or interruptions.
- Maintainability: Code should be modular, well-documented, and version-controlled for ease of updates and debugging.

8.3. System Requirements

- **Hardware Requirements:**
  - Minimum 16 GB RAM system for local development and model execution.
  - Microphone/headset for student speech input (web-based access).
  - Hosting environment or cloud-based server for deployment.
- **Software Requirements:**
  - Frontend: React.js (for quiz interface, dashboards).
  - Backend: FastAPI (Python) for API services and ML integration.
  - Database: PostgreSQL for storing quiz data, results, and user profiles.
  - Libraries/Models:
  - ASR: Whisper / wav2vec 2.0
  - NLP: spaCy, BERT, sentence-transformers
  - Visualization: Recharts or Chart.js for progress display.

8.4. User Requirements

1. Access & Consent

- Students must be able to register/sign in and accept clear consent/privacy terms before any data capture.

2. Assessment Experience

- Students must be guided through a simple flow, with mic permission/status indicators and the ability to re-record before submission.

3. Results & Transparency

- After submission, students must receive a results page within a reasonable time showing three scores, Communication, Problem-Solving, Critical Thinking with brief, plain-English explanations of how each was assessed.

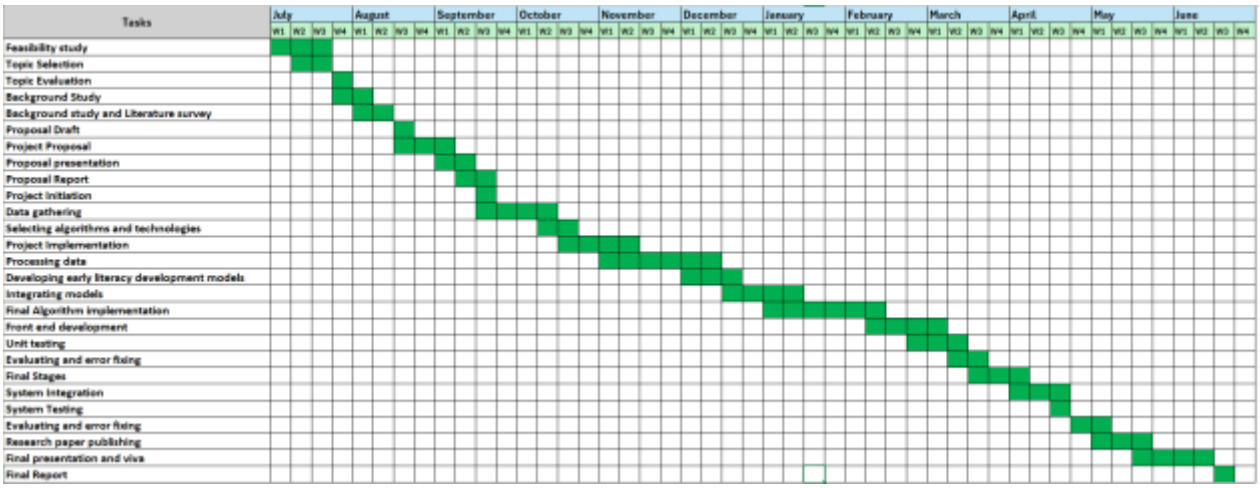
4. personalized Recommendations

- Students must receive 2–3 targeted resources per weak area (course/blog/article/task) and be able to mark items as viewed/completed and rate them as helpful/not helpful.

5. Progress & Privacy

- Students must have a personal dashboard showing history and trends for each skill, with all data kept private to the student. and a clear option to request deletion of their recordings/attempts.

8.5. GANTT Chart



8.6. Work Breakdown Structure

1. **Initiation & Planning**
   - Confirm scope, skills, and success criteria
   - Stakeholder mapping & ethics approvals (consent, privacy)
   - Project plan & risk register (timeline, dependencies)
2. **Data Design & Collection**

- Instrument desig
  - Speech prompts (communication)
  - MCQs (problem-solving, critical thinking)
  - Consent forms & participant info sheets
- Data collection setup
  - Lightweight web form for recordings & quizzes
  - Storage schemas (audio, transcripts, quiz results)
- Participant recruitment & sessions (50–100 learners)
- Data preprocessing
  - Anonymization & quality checks
  - Transcription baselines (for spot checks)

3. **System Design & Architecture**

- Data model & API contracts
- Evaluation rubrics (feature-to-score mapping)
- Recommendation rules (thresholds, skill to resource mapping\

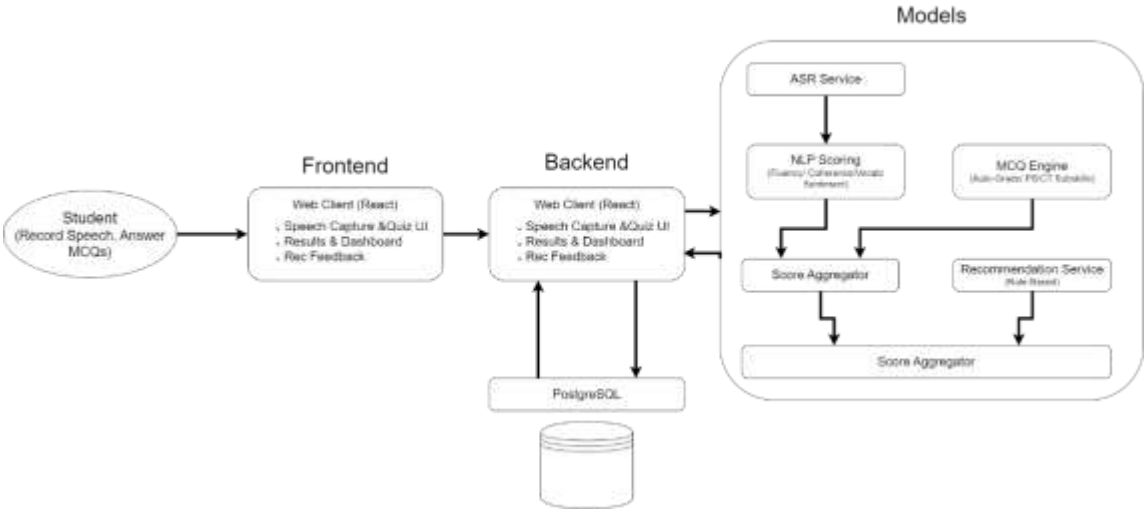


Fig.1. System Design

4. Model Development (AI/ML)

- ASR
  - Whisper integration & audio handling
  - WER baseline & noise handling
- NLP scoring (communication)
  - Feature extraction: fluency, coherence, vocabulary, sentiment
  - Fine-tuning / calibration against small gold labels
  - Scoring aggregation & normalization
- Quiz scoring & analytics
  - MCQ engine (timing, auto-grade)
  - Skill-dimensional mapping (PS/CT subskills)
- Recommendation model
  - Online course recommendation

5. Application Development

- Backend (FastAPI)
  - Auth & user profiles
  - ASR/NLP inference endpoints
  - Quiz & scoring services

    - Recommendation service (rule-based)
- Frontend (React)
    - Speech capture & quiz UI
    - Student dashboard (radar, trend lines)
    - Recommendation view & feedback capture
- Database (PostgreSQL)
    - Schemas (users, attempts, scores, recs)
    - Migrations & indexes
- DevOps basics
    - Env & secrets management
    - Containerization (optional) & deployment script

6. Integration & Testing

- Module integration (end-to-end flows)
- Functional tests (ASR to NLP to Scores to Recs to Dashboard)
- Non-functional tests (latency, basic load, reliability)
- Pilot run (small cohort), defect triage & fixes

7. Evaluation & Analysis

- Technical metrics (WER, scoring consistency vs. human)
- User study (usability, clarity, usefulness)
- Pre/post comparison (skill gains)
- Results synthesis & limitations

8. Documentation & Delivery

- Research report (method, results, discussion)
- User guide (how to use the system)
- Demo script & presentation deck
- Final submission packaging (code, data cards)

9. Project Management & Quality

- Weekly progress tracking & minutes

- Version control & code reviews (solo checklist)
- Quality gates (acceptance criteria per milestone)

9. BUDGET ALLOCATION

- Expert Consultation: To ensure methodological soundness and contextual relevance of soft-skill evaluation and feedback
- Data Collection & Annotation: For building a high-quality prototype dataset of speech recordings and MCQ responses. Participant recruitment, Session logistics, scheduling, and supervision, Preprocessing, Light **gold-labeling** to calibrate NLP sub-scores
- Software, Tools, and Infrastructure: Core development and hosting stack.
- Like ASR/NLP, Application, Cloud/Compute, and Dev tooling
- System Development & Integration
    - Engineering effort to build and connect all modules.
    - Speech capture UI and recording flow.
    - ASR pipeline (English) and transcript storage.
    - NLP scoring for communication (fluency, coherence, vocabulary, sentiment) and aggregation to 0–100.
    - MCQ engine for problem-solving and critical thinking (item bank, timing, auto-grade)
    - Rule-based recommendation service mapped to score thresholds, using a curated resource catalog.
    - Student dashboard (scores, trends, profile, recommendation history).
    - Admin minisuite for prompts, MCQs, and resource catalog management.
    - Security, logging, and basic analytics.
- **Testing and Validation:** Ensuring the system is accurate, usable, and reliable by doing Functional tests, Technical Metrics, Usability testing, accessibility and cross browser testing and performance and basic testing.
- **Operational Costs**: Paying for long-term system maintenance, technical assistance, documentation, security compliance, and administrative costs

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